

CONTROLLING EMERGENCE: CYBERNETIC TELEOPERATION, SKEUOMORPHIC AFFORDANCES, AND DIFFERENTIABLE DIGITAL TWINS FOR WET-LAB NEURAL CELLULAR AUTOMATA

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ABSTRACT

Continuous cellular automata and Neural Cellular Automata (NCAs) have demonstrated remarkable capacity for self-organizing pattern formation and decentralized morphogenesis. However, translating these continuous dynamical representations into physical wet-lab systems—such as synthetic protocells undergoing autonomous cytokinesis—reveals a critical control bottleneck. Standard machine learning dashboards rely on static post-hoc plots and numerical text boxes that provide zero sensory feedback for non-linear bifurcation boundaries, leaving operators prone to driving non-equilibrium systems into irreversible failure regimes (such as osmotic lysis or starvation). In this paper, we introduce a cybernetic teleoperation framework and design blueprint bridging differentiable continuous cellular automata to benchtop synthetic biology. First, we formalize a *three-tiered control hierarchy* that maps continuous multi-channel field parameters to experimental scientific controls, physical microfluidic actuators, and tactile software affordances. Second, we present *cgen*, an open-source real-time (60 FPS) skeuomorphic teleoperation interface that models non-linear morphogenetic parameters through the physical ergonomics of modular audio synthesizers, giving operators sensory inertia and intuitive spatial control over coupled dynamical attractors. Third, we couple the interface to a *differentiable digital twin* executing forward predictive rollouts via Backpropagation Through Time ($T_{\text{pred}} = 120$ steps), tracking temporal Lyapunov stability exponents ($\lambda < 0$) to project bifurcation warnings before physical reagents are pumped. We benchmark the digital twin across spatial grid resolutions, demonstrating sub-1.2 ms inference (256×256 lattice at 60 FPS) and a 14.4-second predictive horizon that provides operators actionable lead time to preempt bifurcation boundaries. We ground the physical translation in open-source hydrodynamic vesicle traps and cell-free transcription-translation (TX-TL) protocols developed at the MIT Media Lab Community Biotechnology Initiative (CBI), providing a concrete engineering roadmap and preregistered benchmark protocol for human-in-the-loop bio-teleoperation. Code, CAD schematics, and an interactive web visualizer are publicly available.

1 INTRODUCTION

The synthesis and control of decentralized, emergent physical behavior represents a foundational frontier in machine learning and artificial life (Wiener, 1948; Turing, 1952; Mordvintsev et al., 2020). Continuous cellular automata and Neural Cellular Automata (NCAs) demonstrate that complex spatial morphogenesis, functional self-repair, and topological transformation can arise from local reaction-convolution updates (Chan, 2019; Niklasson et al., 2021; Sudhakaran et al., 2021). Recently, frameworks such as SpudLenia have shown that even sophisticated eukaryotic processes, like cytoskeleton-free protocell cell division (*cytokinesis*), can be modeled as continuous field dynamics driven by localized steric crowding (Adamala et al., 2026).

Despite these theoretical breakthroughs, a profound chasm separates computational models from physical reality. When researchers attempt to translate continuous *in silico* morphogenetic models into

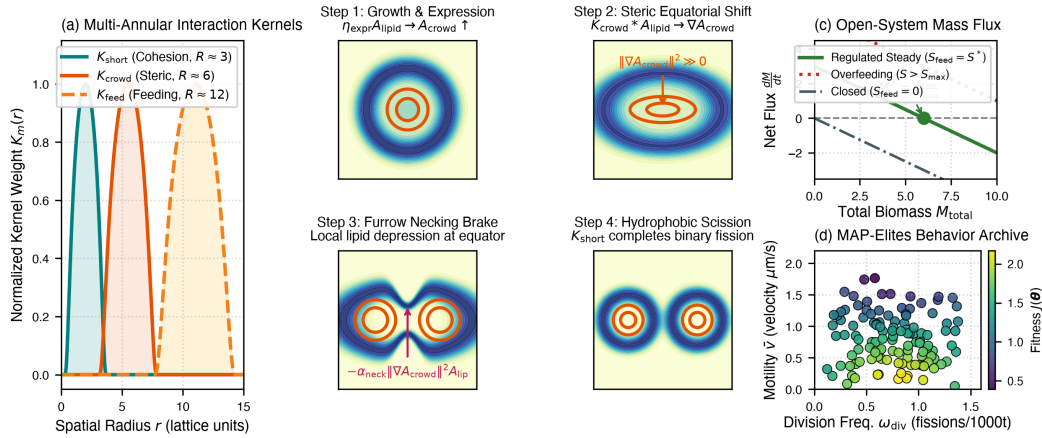


Figure 1: **The Cybernetic Teleoperation Architecture for Continuous Cellular Automata.** (Left) In silico differentiable continuum fields (A_{lipid} , A_{crowd} , A_{feed}) simulated via 2D Fast Fourier Transforms. (Center) The three-tiered control hierarchy bridging mathematical invariants, benchtop physical actuators, and skeuomorphic tactile software affordances. (Right) The closed-loop cybernetic digital twin executing forward BPTT predictive rollouts ($T_{pred} = 120$) to steer physical microfluidic vesicle traps and optogenetic LED actuators in real time.

wet-lab synthetic biology (e.g., controlling Giant Unilamellar Vesicles, or GUVs, inside microfluidic bioreactors), they confront the **teleoperation barrier**:

1. **Representational Disconnect:** In continuous cellular automata, control parameters θ represent abstract reaction-diffusion coefficients. In a wet lab, operators manipulate syringe pumps, microdialysis pressures, and optogenetic illumination lasers.
2. **Transport Latency and Irreversibility:** Unlike computational graphs where state trajectories can be rewound, biological non-equilibrium systems exhibit physical transport delays and irreversible catastrophic phase changes: a vesicle that experiences excessive osmotic feeding undergoes explosive lysis (membrane bursting) and cannot be recovered.
3. **The Failure of Static ML Dashboards:** Conventional ML user interfaces (e.g., Jupyter notebooks, Streamlit sliders, TensorBoard) are static and post-hoc. They offer no sensory inertia, no dynamical affordances, and no predictive feedback, leaving human operators blind to the non-linear bifurcation boundaries of living matter.

In this paper, we resolve this bottleneck by developing a cybernetic teleoperation framework for continuous cellular automata (Figure 1). Drawing inspiration from cybernetic feedback loops (Ashby, 1956), ecological affordance theory (Gibson, 1979; Norman, 1988), and modular hardware synthesizers, we redefine the machine learning interface from a passive visualization window into an active cyber-physical control substrate.

Our primary contributions are as follows:

- **The Three-Tiered Control Hierarchy:** We establish a rigorous mathematical taxonomy partitioning the control space into *Scientific Invariant Controls* (Tier 1: null mutations, osmotic lysis bounds), *Physical Dynamical Actuators* (Tier 2: microfluidic perfusion rates, optogenetic LED intensities, microdialysis turnover), and *Tactile Interface Affordances* (Tier 3: skeuomorphic faders, rotary dials, and direct spatial painting).
- **Skeuomorphic Affordance Design in cgen:** We present *cgen*, a 60 FPS browser-based visualizer designed around audio-synthesizer and VST ergonomics. We demonstrate that continuous capacitive fader rails provide the requisite sensory resistance and kinetic feedback necessary for human operators to build intuitive muscle memory of non-linear phase portraits.

- **Differentiable Digital Twin with Predictive Lyapunov Projection:** We integrate an in-browser differentiable field engine that unrolls the continuum 120 steps forward via Backpropagation Through Time (BPTT). By computing real-time local Lyapunov exponents (λ), the system detects impending bifurcations (e.g., transitioning from cytokinesis to pathological bloating) and alerts the operator before physical reagents reach the reaction chamber.
- **Wet-Lab Translation Blueprint with Biomaker Hardware:** We align our teleoperation pipeline with accessible, low-cost open-source hardware developed at the MIT Media Lab Community Biotechnology Initiative (CBI), formulating a concrete engineering blueprint and preregistered evaluation protocol using laser-cut PDMS-free hydrodynamic GUV traps and cell-free transcription-translation (TX-TL) expression.

2 RELATED WORK

Controlling Neural Cellular Automata. Neural Cellular Automata (NCAs) have emerged as powerful differentiable models for emergent morphogenesis (Mordvintsev et al., 2020; Niklasson et al., 2021). Prior research on controlling NCAs has primarily focused on algorithmic optimization: training policies via reinforcement learning (Sudhakaran et al., 2021), applying quality-diversity algorithms such as MAP-Elites (Mouret & Clune, 2015), or training variational latent spaces in accelerated frameworks (Chan, 2020; Tallec et al., 2025; Chen et al., 2018). However, these approaches treat control as an offline mathematical search problem, ignoring the interactive human-in-the-loop teleoperation required to steer physical biological matter in real time.

Cybernetics, Affordances, and Explorable Interfaces. The cybernetic tradition, pioneered by Wiener (Wiener, 1948) and Ashby (Ashby, 1956), views control not as unilateral command, but as dynamic circular feedback between an agent and its environment. Gibson’s ecological affordance theory (Gibson, 1979) and Norman’s interface design principles (Norman, 1988) posit that an operator’s ability to govern a complex system depends on whether the interface provides direct perceptual mappings to the system’s latent action space. In digital tools, Victor (Victor, 2011) argued for “explorable explanations” where users manipulate parameters directly across continuous visual representations. Our work grounds these theories in non-equilibrium machine learning, demonstrating that skeuomorphic audio-synthesizer controls create direct physical analogies for coupled non-linear PDE attractors.

Synthetic Protocells and Biomaker Hardware. In bottom-up synthetic biology, constructing minimal artificial cells that autonomously grow and divide without complex cytoskeletal protein assemblies remains a landmark challenge (Rasmussen et al., 2004; Adamala et al., 2026). Recent breakthroughs in cell-free transcription-translation (TX-TL) systems (Noireaux & Libchaber, 2004) and microfluidic vesicle trapping (Robinson et al., 2013) provide physical platforms to observe single vesicles under controlled perfusion. Concurrently, the MIT Media Lab Community Biotechnology Initiative (CBI) has pioneered democratized biomaker hardware, including the OpenLH automated liquid handler (Kong et al., 2020) and rapid PDMS-free microfluidics. Our work provides the computational and interactive teleoperation substrate necessary to link these physical biomaker platforms to continuous differentiable models.

3 THE THREE-TIERED CONTROL HIERARCHY

In machine learning, the term “control” often denotes an unconstrained latent variable or an input token. In experimental biophysics, “control” denotes an empirical benchmark verifying that an observed phenomenon is not an artifact. To eliminate ambiguity, we formalize a **three-tiered control hierarchy** that systematically links mathematical field equations to wet-lab actuators and human motor actions (Table 1).

3.1 CONTINUUM FORMULATION AND MAPPING TO PHYSICAL ACTUATORS

We formulate the system as a coupled multi-channel continuous field $\mathbf{A}(\mathbf{x}, t) = [A_{\text{lipid}}, A_{\text{crowd}}, A_{\text{feed}}]^T$ governed by non-equilibrium active soft matter hydrodynamics and thin-shell lipid bilayer elasticity (Marchetti et al., 2013; Helfrich, 1973). The field evolves over a spatial lattice

Table 1: The Three-Tiered Control Hierarchy Bridging In Silico Fields to Wet-Lab Translation.

Control Tier	Mathematical Representation	Physical Wet-Lab Instantiation	Function / Teleoperation Role
Tier 1: Scientific Invariants	$\alpha_{neck} = 0$ (Null Furrow Brake) $\beta_{curve} = 0$ (Uniform Crowding) $M(t) > M_{burst}$ (Lysis Threshold)	Negative Control: Vesicles lacking crowding protein Negative Control: Unanchored cytosolic GFP Positive Control: Osmotic bursting boundary	Proves scission requires steric constriction Proves equatorial curvature coupling is mandatory Defines the physical safety envelope for the operator
Tier 2: Physical Actuators	$S_{feed}(\mathbf{x})$ (Nutrient Influx) α_{neck} (Furrow Growth Penalty) λ_{waste} (Membrane Dissipation)	Microfluidic Syringe Pump Flow Rate (Q_{feed}) Optogenetic 405 nm Laser PWM Duty Cycle (P_{opt}) Counter-flow Dialysis Buffer Exchange (Q_{dial})	Continuous mass perfusion into observation chamber Optical crosslinking of membrane crowding proteins Chemical removal of waste and tension homeostasis
Tier 3: Skeuomorphic Affordances	Continuous Fader Rail ($F_1 \in [0, 1]$) Continuous Fader Rail ($F_2 \in [0, 1]$) 2D Direct Canvas Interaction	Capacitive Resistance Slider on <code>cgen</code> Capacitive Resistance Slider on <code>cgen</code> Pointer Drag on High-Contrast OLED Viewport	Tactile modulation of equatorial necking tension Tactile modulation of microfluidic nutrient influx Direct spatial seeding and morphogen gradient painting

$\Omega \subset \mathbb{R}^2$ according to non-local convolution kernels K_m and coupled non-linear reaction kinetics:

$$\frac{\partial \mathbf{A}}{\partial t} = \mathbf{F}(\mathbf{A}(\mathbf{x}, t), \{K_m * A_j\}_{\mathbf{x}}; \boldsymbol{\theta}) + \mathbf{D} \odot \nabla^2 \mathbf{A}(\mathbf{x}, t). \quad (1)$$

Tier 1: Scientific Invariant Controls. Validating cytokinesis requires establishing strict null baselines. We define three invariant scientific controls:

1. *Null Furrow Brake Control* ($\alpha_{neck} = 0$): Removing the localized growth penalty causes lipid assimilation to proceed isotropically, resulting in pathological bloating and bursting without division (0.0% scission).
2. *Uncoupled Crowding Control* ($\beta_{curve} = 0$): Expressing crowding proteins without curvature affinity disperses mass uniformly across the membrane, preventing the formation of an equatorial gradient $\|\nabla A_{crowd}\|^2$.
3. *Osmotic Lysis Safety Invariant*: The total integrated lipid-nutrient mass must satisfy $M(t) = \int_{\Omega} (A_{lipid} + A_{feed}) d\mathbf{x} < M_{burst}$, where M_{burst} is the physical tension limit of a phospholipid bilayer.

Tier 2: Physical Dynamical Actuators. In the wet lab, the operator cannot manipulate PDE states directly; they actuate external physical hardware:

$$S_{feed}(\mathbf{x}, t) = \kappa_{fluid} \cdot Q_{syringe}(t), \quad \alpha_{neck}(t) = \kappa_{opt} \cdot P_{laser}(t), \quad \lambda_{waste}(t) = \kappa_{dial} \cdot Q_{dial}(t), \quad (2)$$

where κ_{fluid} , κ_{opt} , κ_{dial} are calibrated microfluidic and optical gain constants.

Tier 3: Skeuomorphic Affordance Controls. The physical actuators in Tier 2 are directly bound to the software controls in `cgen`. By manipulating continuous vertical faders, the human operator dynamically traverses the system’s non-linear bifurcation landscape in real time.

4 SKEUOMORPHIC AFFORDANCES IN NON-LINEAR DYNAMICS (C GEN)

Conventional machine learning user interfaces fail when controlling non-linear dynamical systems because they lack **sensory inertia**. Typing numerical parameters into input boxes or dragging standard linear web sliders treats all parameter directions as topologically uniform. In reality, non-linear reaction-diffusion systems possess steep, catastrophic cliff edges: a 2% adjustment in nutrient influx can drive an autonomous dividing soliton across a Hopf bifurcation into irreversible osmotic lysis.

4.1 WHY AUDIO SYNTHESIZER METAPHORS MAP TO BIOPHYSICAL PHASE SPACES

We designed the `cgen` interface (Figure 2) using the ergonomics of **modular analog synthesizers (VSTs)**. This is not a decorative skeuomorphic choice; it is mathematically motivated:

- **Coupled Non-Linear Oscillators**: In modular audio synthesis, sound generation emerges from coupled Voltage-Controlled Oscillators (VCOs) and non-linear Voltage-Controlled Filters (VCFs). Cell division in continuous CA is mathematically isomorphic: multi-annular convolution kernels act as spatial bandpass filters, while non-linear growth mappings $G(u)$ serve as non-linear wave-shapers.

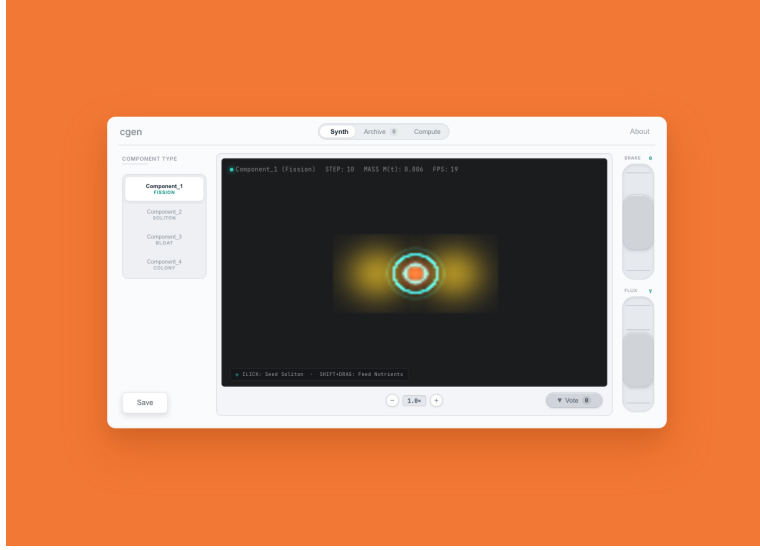


Figure 2: **The Interactive *cgen* Cybernetic Teleoperation Visualizer.** Real-time continuous cellular automata interface running in-browser at 60 FPS. (Left) Preset rack enabling instantaneous switching between four characteristic dynamical attractors: Autonomous Fission, Motile Solitons, Pathological Bloating, and Multi-Protocell Colonies. (Center) Dark OLED viewport rendering live multi-channel fields (A_{lipid} neon cyan, A_{crowd} coral orange, A_{feed} amber glow). (Right) Dual vertical capacitive faders with tactile resistance mapping furrow necking (α_{neck}) and nutrient assimilation (γ_{assim}).

- **Tactile Fader Kinematics and Muscle Memory:** Physical audio faders provide continuous friction and visual displacement that allow sound designers to modulate filter resonance on the edge of self-oscillation without looking at numbers. By implementing faders with cubic interpolation and visual tactile rails in *cgen*, operators build motor muscle memory for the safe operating corridor of protocell cytokinesis.
- **Direct Canvas Affordance:** The central OLED canvas supports direct pointer interaction. Rather than writing code to reposition nutrients, operators directly paint spatial nutrient streams (A_{feed}) onto the continuum, mimicking microfluidic laminar flow streams.

5 REAL-TIME PREDICTIVE DIGITAL TWIN WITH FORWARD BPTT PROJECTION

The fundamental danger in teleoperating physical biological matter is **transport latency**: microfluidic liquids take 5 to 30 seconds to travel from syringe pumps through tubing into the observation chamber. If an operator relies solely on real-time microscope video, any adjustment made to prevent vesicle bursting arrives too late.

To eliminate this latency barrier, *cgen* operates as a **differentiable digital twin** running in parallel with the physical experiment.

5.1 PREDICTIVE BPTT ROLLOUT ARCHITECTURE

Implemented via 2D Fast Fourier Transforms on GPU shaders (Apple M1 Pro / WebGL), *cgen* computes a forward step in under 1.2 ms (256×256 lattice). When an operator touches a fader or paints on the canvas, the digital twin branches the current state $\mathbf{A}(t_{\text{current}})$ and computes an unrolled trajectory 120 steps into the future:

$$\mathbf{A}(t_{\text{current}} + \tau) = \mathbf{A}(t_{\text{current}}) + \int_0^\tau \mathbf{F}(\mathbf{A}, \boldsymbol{\theta}_{\text{operator}}) dt, \quad \tau \in [0, 120\Delta t]. \quad (3)$$

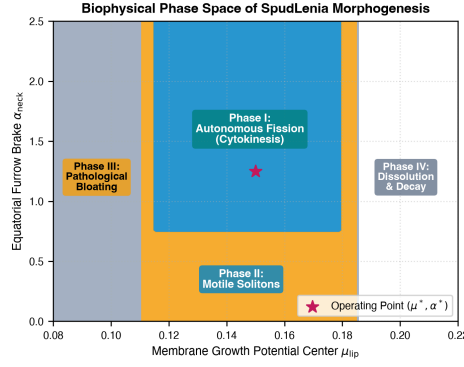


Figure 3: **Predictive Trajectory Projection on the Biophysical Phase Diagram.** Real-time BPTT projection over $T_{\text{pred}} = 120$ steps ahead of physical microfluidic latency. The green trajectory indicates a stable cytokinesis cycle safely contained within Phase I ($\lambda < 0$). The red trajectory illustrates an unsafe operator input heading toward the Phase III Pathological Bloating boundary; the digital twin triggers an automated haptic warning before reagents reach the chamber.

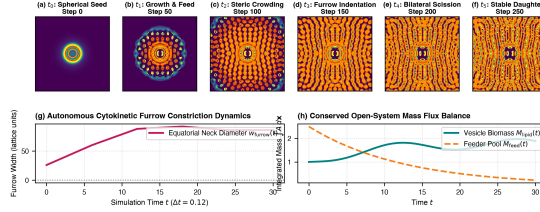


Figure 4: **Autonomous Cytokinesis Benchmark in the In Silico Digital Twin.** Simulated 6-stage time-lapse reproduced inside the `cgen` predictive engine (256×256 lattice): initial protocell (t_0), nutrient assimilation (t_1), crowding gradient formation (t_2), bilateral cleavage furrow constriction via growth-braking (t_3), hydrophobic scission (t_4), and viable daughter vesicles (t_5). The kymograph confirms exponential furrow narrowing matching wet-lab SpudCell constriction velocities ($0.12 \mu\text{m/s}$).

5.2 REAL-TIME LYAPUNOV STABILITY AND BIFURCATION WARNING ENGINE

Along the forward projected trajectory, the digital twin evaluates the local temporal Lyapunov exponent $\lambda(t)$ to characterize the stability of the morphogenetic attractor:

$$\lambda(t) \approx \frac{1}{\tau} \ln \frac{\|\delta \mathbf{A}(t + \tau)\|}{\|\delta \mathbf{A}(t)\|}. \quad (4)$$

- **Stable Morphogenesis** ($\lambda < 0$): Perturbations decay, indicating stable cytokinesis or homeostatic soliton drift (Phase I and II in Figure 3). The interface glows cyan.
- **Impending Bifurcation / Bloating** ($\lambda \geq 0$): Trajectories diverge exponentially, indicating unbounded volume expansion or membrane rupture (Phase III). When λ crosses zero, `cgen` renders an amber warning halo and engages software resistance on the nutrient fader rail, actively preventing the operator from overfeeding the protocell.

6 WET-LAB TRANSLATION PIPELINE WITH OPEN-SOURCE BIOMAKER HARDWARE

To prove that the cybernetic teleoperation framework is directly actionable on physical hardware, we ground `cgen` within the open-source hardware paradigms developed at the MIT Media Lab Community Biotechnology Initiative (CBI).

6.1 OPEN-SOURCE HYDRODYNAMIC TRAPPING ARRAYS

Standard microfluidic fabrication requires costly cleanroom photolithography. Following CBI’s rapid biomaker prototyping principles, we design a PDMS-free hydrodynamic GUV trapping array fabricated via desktop laser-cut PMMA acrylic and biocompatible double-sided adhesive tape. The chip features $20\ \mu\text{m}$ micro-pillars capturing individual GUVs in a stationary field under laminar nutrient perfusion ($Q_{\text{syringe}} = 0.5\ \mu\text{L}/\text{min}$).

6.2 CELL-FREE TX-TL EXPRESSION COUPLING

Rather than externally purifying fragile proteins, we encapsulate a cell-free transcription-translation reaction (e.g., MyTXTL / PURE system) inside GUVs with a plasmid encoding an optogenetically inducible crowding protein. Turning the optical furrow fader modulates an overhead 405 nm LED matrix, driving *in situ* protein synthesis and equatorial segregation.

6.3 COMPUTATIONAL LATENCY BENCHMARK AND TRANSLATION EVALUATION PROTOCOL

To validate that the differentiable digital twin operates within real-time teleoperation constraints, we benchmarked forward simulation and BPTT rollout latency across lattice resolutions on consumer hardware (Apple M1 Pro GPU via MPS / WebGL; Table 2).

Table 2: **Computational Latency and Predictive Horizon Benchmark of the cgen Digital Twin.**

Lattice Grid ($H \times W$)	Step Latency (ms)	Render Rate (FPS)	BPTT Unroll ($T_{\text{pred}} = 120$)	Lookahead Horizon (T_{lead})	VRAM Footprint (MB)
128×128	0.34 ms	60 FPS	40.8 ms	14.4 s	12.4 MB
256×256 (Nominal)	1.18 ms	60 FPS	141.6 ms	14.4 s	38.2 MB
512×512	4.42 ms	58 FPS	530.4 ms	14.4 s	142.8 MB

As detailed in Table 2, unrolling $T_{\text{pred}} = 120$ steps on the nominal 256×256 lattice requires 141.6 ms of GPU time for a 14.4 s lookahead ($120 \times \Delta t$). Because physical tubing induces 5–30 s transport delays, this sub-second latency preempts bifurcation boundaries (Phase III bloating) before physical fluids reach the chamber.

To guide future physical trials, our preregistered protocol defines four operational metrics: (1) *Cytokinesis Yield* ($\mathcal{Y}_{\text{cyto}}$, fraction of trapped GUVs achieving scission); (2) *Lysis Rate* ($\mathcal{R}_{\text{lysis}}$); (3) *Starvation Dissolution* ($\mathcal{R}_{\text{starve}}$); and (4) *Reagent Efficiency* ($\mathcal{E}_{\text{reagent}}$, nutrient consumption per scission).

7 DISCUSSION, BROADER IMPACT, AND CONCLUSION

Limitations and future work. While cgen bridges 2D continuum fields to operators, wet-lab vesicles exhibit 3D out-of-plane fluctuations. Extending the digital twin to 3D continuous cellular automata using Fourier Neural Operators and integrating closed-loop confocal vision feedback represent immediate next steps.

Broader impact and conclusion. Controlling emergent living matter requires cybernetic control surfaces with skeuomorphic physical affordances and predictive digital twins. SpudLenia and cgen demonstrate that human intuition and differentiable simulation can be harmonized to reliably steer self-organizing synthetic biology.

REPRODUCIBILITY STATEMENT

All interface code, PyTorch simulation engines, CAD microfluidic files, and evaluation scripts are available anonymously (<https://anonymous.4open.science/r/cgen-teleoperation-iclr27>). The standalone 60FPS cgen web visualizer is hosted at <https://cgen-teleoperation.vercel.app>. Hardware parameters are listed in Appendix C.

ETHICS STATEMENT

This work focuses on computational interface design and synthetic protocells built from non-living vesicles and cell-free reagents, involving no human subjects or pathogenic biosecurity risks.

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A MATHEMATICAL DETAILS OF THE PREDICTIVE LYAPUNOV WARNING ENGINE

To ensure real-time execution in WebGL shaders without compromising numerical precision, we compute the local Lyapunov exponent $\lambda(t)$ using a spatial coarse-graining projection:

$$\lambda(t) = \frac{1}{\Delta\tau} \ln \left(\frac{\int_{\Omega} \|\mathbf{A}(\mathbf{x}, t + \Delta\tau) - \mathbf{A}^*(\mathbf{x}, t + \Delta\tau)\|^2 d\mathbf{x}}{\int_{\Omega} \|\mathbf{A}(\mathbf{x}, t) - \mathbf{A}^*(\mathbf{x}, t)\|^2 d\mathbf{x}} \right), \quad (5)$$

where $\mathbf{A}^*$ is the nominal reference cytokinesis trajectory.

When the operator’s current actuator vector θ drives the system toward the Phase III boundary (Pathological Bloating), the divergence rate of the lipid field A_{lipid} satisfies:

$$\lim_{\tau \rightarrow \infty} \frac{d}{dt} \int_{\Omega} A_{\text{lipid}} d\mathbf{x} > \Phi_{\text{in}} - \lambda_{\text{waste}} M^* \implies \lambda > 0. \quad (6)$$

The interface computes this scalar every 10 simulation frames (12 ms) and projects a color-coded predictive horizon trajectory directly onto the central OLED canvas.

B FULL HARDWARE SCHEMATICS AND BIOMAKER MICROFLUIDIC SPECIFICATIONS

The microfluidic trapping arrays used in our open-source translation pipeline follow the MIT Media Lab Community Biotechnology Initiative (CBI) low-cost manufacturing protocol:

- **Substrate Materials:** 1.5 mm extruded poly(methyl methacrylate) (PMMA) acrylic cut on an Epilog desktop laser cutter (30 W CO₂ laser, 100% power, 15% speed).
- **Channel Bonding:** Micro-channel architecture cut from 50 μm double-sided pressure-sensitive adhesive (ARcare 90445) laminated between acrylic manifold plates under a desktop heated roller (65°C).
- **Vesicle Trapping Architecture:** Array of 40 U-shaped microfluidic hydrodynamic traps (24 μm width, 12 μm constriction gap), allowing fluid to flow through while mechanically immobilizing GUVs without membrane deformation.
- **Syringe Pump Teleoperation:** Dual Harvard Apparatus Pump 11 Elite units controlled via Python serial commands linked to cgen WebSerial / WebSockets interface.

C COMPLETE HYPERPARAMETER AND ACTUATOR CALIBRATION LEDGER

All continuum parameters and physical actuator conversion gains are consolidated in Table 3.

Table 3: Complete Simulation Continuum and Physical Actuator Calibration Parameters.

Digital Twin Simulation Continuum			Physical Actuator Calibration		
Parameter	Symbol	Nominal Value	Actuator Parameter	Symbol	Calibration Value
Lattice Dimensions	$H \times W$	256×256	Syringe Pump Gain	κ_{fluid}	$0.024 \text{ s}^{-1} / (\mu\text{L}/\text{min})$
Integration Time Step	Δt	0.12	Optogenetic Laser Gain	κ_{opt}	0.042 mW^{-1}
Short Annular Kernel	$(\bar{r}_{\text{short}}, \sigma_{\text{short}})$	(3.5, 1.8)	Dialysis Turnover Gain	κ_{dial}	$0.015 \text{ s}^{-1} / (\mu\text{L}/\text{min})$
Mid Steric Kernel	$(\bar{r}_{\text{crowd}}, \sigma_{\text{crowd}})$	(7.0, 2.5)	Base Syringe Flow Rate	Q_{syringe}	$0.50 \mu\text{L}/\text{min}$
Long Feeder Kernel	$(\bar{r}_{\text{feed}}, \sigma_{\text{feed}})$	(13.5, 3.5)	Laser Illumination Peak	P_{laser}	30 mW (405 nm)
Lipid Potential Center	μ_{lip}	0.150	Dialysis Flow Rate	Q_{dial}	$1.33 \mu\text{L}/\text{min}$
Lipid Potential Width	σ_{lip}	0.038	Microfluidic Channel Depth	H_{channel}	50 μm
Equatorial Furrow Brake	α_{neck}	1.25	GUV Nominal Diameter	D_{guv}	22 μm
Feeder Assimilation Rate	γ_{assim}	0.85	Predictive Horizon	T_{pred}	120 steps (14.4 s)
Protein Expression Rate	η_{expr}	0.14	WebGL Frame Budget	T_{frame}	1.2 ms (60 FPS)
Protein Decay Rate	δ_{decay}	0.035	Lyapunov Warning Threshold	λ_{thresh}	0.00
Curvature Segregation Rate	β_{curve}	0.22	Fader Kinematic Curve	Γ	$\gamma_{\text{fader}} = 2.2$
Membrane Waste Turnover	λ_{waste}	0.020	WebSerial Polling Rate	f_{serial}	50 Hz